SWE 432 – Fall 2026

Assignment 3: Data Analysis with PHP

Purpose

In this assignment, you will download a public dataset from the web and write a simple data analysis application in PHP to answer ten questions about your dataset. It's the same task you did on the front-end with JavaScript, only this time you will do it on the server-side with PHP.

Instructions

Step 1: Download a JSON dataset

First, you will download a JSON dataset from a public source like Open Data DC, Virginia Open Data, or any other public repository. You are free to choose whatever data source you'd like. The dataset you select must contain at least 1000 records in JSON format.

Step 2: List the questions you will answer about your dataset

Now that you've found a dataset, what insights can you extract from this dataset? In this step, you will write a list of ten (10) questions about your dataset. Each question should describe a statistic to compute from your dataset. For example, if your dataset is city demographic data, you might have questions like the following:

- What is the average age of residents?
- What is the fastest growing city?
- What is the median population density?
- What is the city with the highest population density?

Note: If any of the questions you picked for HW2 were not adequate (check your HW2 grading and feedback), you must change them to comply with the requirements of the assignment.

Step 3: Implement a basic webpage

Create a basic webpage that has:

- A form (method is post) containing an input element (type is file) that the user can use to upload a `.json` file, and a submit button
- Elements for displaying the 10 questions and the respective statistics (these will be generated by your PHP program when it responds to the upload)
- An element for displaying additional information regarding the data processing

Step 4: Implement the PHP logic

Create PHP code to compute the answers to your questions using the JSON dataset. When the user uploads the `.json` file, the PHP processes the data, calculates the statistics for each question, and generates the HTML response. Do not store any data in the code; the code must read all the data from the uploaded file.

Your program must use all of the following PHP features:

- Superglobals
- htmlspecialchars() to prevent XSS
- Strings (including variable interpolation in double-quoted strings)
- Arrays (including use of array_push, array_map, and array_filter)
- Loops (for and foreach are required; while and do-while are optional)
- Functions (including use of default parameter values)
- Class declarations following good OOP principles (individual files, visibility rules, inheritance, etc.)
- Class methods
- Exceptions and errors (if given an invalid file, show an error message on the webpage)

Additionally, your app must display on the webpage (use the extra element mentioned above) the timestamp and the total time it took to process the data.

Step 5: Deploy the app on Render

Follow the instructions provided in class to deploy your app on Render. Once completed, your app must be publicly accessible in a url of the form `https://xxxxxx.onrender.com`

Submission instructions

The submission should consist of the following files:

1. All the PHP and CSS files compressed into a single .zip file
2. A text file containing the onrender.com URL where your app is deployed
3. A text file containing the URL of the dataset
4. A copy of the JSON dataset that was used
5. A text file containing the 10 questions that your program answers
6. A video presentation (as a file, not a link) of no more than 5 minutes where you demonstrate your complete understanding of the PHP code you wrote.
7. A statement about the use of AI. If you did not use AI, explicitly state that. Otherwise, provide a detailed description of what AI was used for.
   NOTE: The use of generative AI is allowed only if: a) you have a complete understanding of all parts of the code generated with AI assistance, and b) if you're asked to, you are able to edit any part of the code without any kind of assistance.

Evaluation Rubric

Criterion	Weight
Statistics: Correct algorithmic logic for the statistics	**20%**
PHP logic: Your code must contain all requested features	**30%**
Code modularity and OOP: One of the best ways to help make your code more maintainable in the long run is to make it modular and follow the OOP principles. Try to achieve low coupling and high cohesion (e.g. each class in its own file, separate functions for reading/computing/rendering, proper visibilities, etc.)	**10%**
Deployment: Your app must be deployed with Render	**10%**
Documentation & Comments: You should document all non-obvious functionality in your code	**10%**
10 valid questions about the dataset: Select meaningful questions that cannot be answered without programming	**5%**
Valid data: Your JSON dataset must have at least 1000 entries and must have enough attributes to be able to form 10 meaningful questions to answer	**5%**
Video presentation: Clear demonstration of the code and explanation of the implementation choices	**10%**